

STATSIG lecture 6

Extended Kalman filter (EKF)

for non-linear systems

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Extended Kalman filter (EKF)

Linear model

$$\begin{aligned}x_k &= Ax_{k-1} + Bu_{k-1} + w_{k-1} && \text{(process) } (n \times 1) \\z_k &= Hx_k + v_k && \text{(observations) } (m \times 1)\end{aligned}$$

Non-linear model

$$\begin{aligned}x_k &= f(x_{k-1}, u_{k-1}, w_{k-1}) && \text{(process) } (n \times 1) \\z_k &= h(x_k, v_k) && \text{(observations) } (m \times 1)\end{aligned}$$

EKF linearizes the non-linear functions around

$$\begin{aligned}\hat{x}_k^- &= f(\hat{x}_{k-1}, u_{k-1}, 0) \\ \hat{z}_k^- &= h(\hat{x}_k^-, 0)\end{aligned}$$

Linearized process model with Jacobians

$$\begin{aligned}(x_k - \hat{x}_k^-) &= A_k(x_{k-1} - \hat{x}_{k-1}) + W_k(w_{k-1} - 0) \\ A_k &= \left\{ \frac{\partial f_i}{\partial x_j} \right\}_{ij|_{\hat{x}_{k-1}, u_{k-1}, 0}}, \quad W_k = \left\{ \frac{\partial f_i}{\partial w_j} \right\}_{ij|_{\hat{x}_{k-1}, u_{k-1}, 0}}\end{aligned}$$

Linearized observation model with Jacobians

$$\begin{aligned}(z_k - \hat{z}_k^-) &= H_k(x_k - \hat{x}_k^-) + V_k(v_k - 0) \\ H_k &= \left\{ \frac{\partial h_i}{\partial x_j} \right\}_{ij|_{\hat{x}_k^-, 0}}, \quad V_k = \left\{ \frac{\partial h_i}{\partial v_j} \right\}_{ij|_{\hat{x}_k^-, 0}}\end{aligned}$$

Special case – additive noise

$$W_k = I_{nn}, \quad V_k = I_{mm}$$

Time line from sample time k-1 → k

k-1

k

Predict step → Update step

Predict step

$$\hat{x}_k^- = f(\hat{x}_{k-1}, u_{k-1}, 0)$$

$$\hat{z}_k^- = h(\hat{x}_k^-, 0)$$

$$P_k^- = A_k P_{k-1} A_k^T + W_k Q_{k-1} W_k^T$$

Update step

$$\hat{x}_k = \hat{x}_k^- + K_k (z_k - \hat{z}_k^-)$$

where Kalman Gain (minimizing MSE)

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + V_k R_k V_k^T)^{-1}$$

$$P_k = (I - K_k H_k) P_k^-$$

$$k \rightarrow k + 1$$

Algorithm

initialize, e.g. $\hat{x}_0 = 0, P_0 = I, k = 1$

iterate: predict, update, predict, update

Limitations

EKF can diverge – best for weakly non-linear systems